16b. Causal Discovery Going Local: GS, HITON, MMHC, BOSS

A consolidated, mathematical lecture companion

English notes based on the lecture slides *Causal Discovery: Going Local* (Prof. Sanghack Lee, GSDS SNU)

The central question is what happens to structure learning when the reasoning is restricted to local objects (Markov blankets, a target's neighborhood, order prefixes) instead of the whole graph.

These notes consolidate the slides, organize four algorithms on a two-dimensional map, give their pseudocode and correctness arguments (from the appendix), and work the BOSS example. Physical PDF page numbers are listed in Section 5.

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1 The map

Purpose. Organize discovery algorithms by signal and reach.

The previous lecture gave the global workhorses: constraint-based (PC, FCI), using CI tests over the whole graph, and score-based (GES, NOTEARS, DAGMA), optimizing one global score. This lecture is not a new family but a direction: *what happens when we go local?* Two knobs organize everything:

- **Signal** (vertical): test independences or optimize a score?
- **Reach** (horizontal): reason over the whole graph or a local object?

	global	local
CI tests	PC, FCI	GS, HITON
CI $\rightarrow$ score		MMHC
score	GES, DAGMA	BOSS

The previous lecture filled the global column; this one fills the local column.

1.1 One primitive: grow–shrink

A recurring two-step move, reused on different local objects:

- **Grow:** add while the candidate still helps. CI version: add Y while $X \not\perp\!\!\!\perp Y \mid \mathbf{S}$; score version: add Y while the score improves.
- **Shrink:** remove what became redundant. CI version: drop Y if $X \perp\!\!\!\perp Y \mid \mathbf{S} \setminus \{Y\}$; score version: drop Y if the score improves.

Algorithm	Local object	Signal
GS	Markov blanket	CI
HITON	one target's parents and children	CI
MMHC	skeleton, then DAG	CI + score
BOSS	order prefix	score

2 Going local on CI

Purpose. Learn Markov blankets and target neighborhoods by CI tests.

2.1 The Markov blanket

For a node X,

$$\mathrm{MB}(X) = \big(\mathrm{pa}(X) \cup \mathrm{ch}(X) \cup \mathrm{pa}(\mathrm{ch}(X))\big) \setminus \{X\},$$

its parents, children and spouses (co-parents of children). Conditioning on $\mathrm{MB}(X)$ makes every other variable irrelevant for X. It is small and local, yet enough to drive skeleton and orientation (Margaritis and Thrun 1999; Margaritis 2003).

2.2 GS: grow–shrink the blanket

Grow (permissive): add any Y with $X \not\perp\!\!\!\perp Y \mid \mathbf{S}$; some additions are real, others artefacts of an incomplete $\mathbf{S}$. **Shrink** (clean up): drop Y once a better separator appears, $X \perp\!\!\!\perp Y \mid \mathbf{S} \setminus \{Y\}$. Example: in the chain $X \to C \to G$, $X \not\perp\!\!\!\perp G$, so grow may add G; once C is in $\mathbf{S}$, $X \perp\!\!\!\perp G \mid C$ and shrink removes G. Then test which blanket members are true neighbors and orient unshielded colliders as in PC.

Algorithm 1 GS-MB(X)

1: $\mathbf{S} \leftarrow \emptyset$
2: **repeat** find $Y \in \mathbf{V} \setminus (\{X\} \cup \mathbf{S})$ with $X \not\perp\!\!\!\perp Y \mid \mathbf{S}$; if found, $\mathbf{S} \leftarrow \mathbf{S} \cup \{Y\}$
3: **until** no such Y exists
4: **repeat** find $Y \in \mathbf{S}$ with $X \perp\!\!\!\perp Y \mid \mathbf{S} \setminus \{Y\}$; if found, $\mathbf{S} \leftarrow \mathbf{S} \setminus \{Y\}$
5: **until** no such Y exists
6: **return S**

Correctness sketch. Under faithfulness, a parent or child Y of X stays dependent on X given any set not containing X, Y, so grow includes it. If Y is a parent of a child of X, conditioning on that child creates dependence, so grow eventually includes Y. Thus after grow $\mathrm{MB}(X) \subseteq \mathbf{S}$. If $W \notin \mathrm{MB}(X)$ but $W \in \mathbf{S}$, the true blanket (itself a separating set) separates X from W; since $\mathrm{MB}(X) \subseteq \mathbf{S} \setminus \{W\}$, shrink removes W via $X \perp\!\!\!\perp W \mid \mathbf{S} \setminus \{W\}$.

Structure induction. (1) Compute $\mathrm{MB}(X)$ for every X.

(2) For each $Y \in \mathrm{MB}(X)$ decide whether it is a direct neighbor: no edge if $\exists \mathbf{S} \subseteq \mathbf{R}$ with $X \perp\!\!\!\perp Y \mid \mathbf{S}$, where $\mathbf{R}$ is the smaller of $\mathrm{MB}(X) \setminus \{Y\}$ and $\mathrm{MB}(Y) \setminus \{X\}$.

(3) Orient unshielded colliders using local tests.

(4) Repair cycles and propagate forced directions.

Randomized GS. Plain GS becomes expensive with large blankets. Randomized GS samples conditioning sets from the smaller blanket instead of enumerating all subsets and combines multiple tests as evidence for edge existence and direction: an anytime-style algorithm (more tests, more stable decisions) with controlled runtime but weaker exactness guarantees.

2.3 HITON: target-focused, not whole-graph

GS computes a blanket for every node to assemble the full network. HITON wants only the local neighborhood of one target T (e.g. feature selection for a disease variable). Its workhorse HITON-PC($\cdot$) returns a node's *parents and children* ("PC" here is not the PC algorithm) by interleaved grow-while-shrink with small conditioning sets, which is sample-efficient. "Focus" means running this subroutine on T and a bounded local ring around it, not on all n variables: local, not literally single-node. Two pieces: HITON-PC (one node's neighbors) and HITON-MB (assembling T's blanket) (Aliferis et al. 2003; 2010).

HITON-MB: two rings, then prune. Example: $P \to T \to C \leftarrow Z$, $C \to G$.

1. Ring 1: PC(T) $\leftarrow$ HITON-PC(T), T's parents and children (P, C).
2. Ring 2 (PC-of-PC): run HITON-PC(Y) for each $Y \in \mathrm{PC}(T)$. A spouse Z is not adjacent to T, so it only surfaces here, as a neighbor of a child.
3. Prune: drop ring-2 candidates still separable from T (e.g. the grandchild G, separated by C). Ring 2 keeps the spouse Z (no set separates it from T once the collider C is conditioned on) and prunes G.

This calls HITON-PC on several nodes, but only on T's local ring. The PC step finds direct

Algorithm 2 HITON-MB(T)

1: PC(T) $\leftarrow$ HITON-PC(T); $\mathbf{C} \leftarrow$ PC(T)
2: **for** each $Y \in$ PC(T) **do** $\mathbf{C} \leftarrow \mathbf{C} \cup$ HITON-PC(Y)
3: **for** each candidate $X \in \mathbf{C} \setminus$ PC(T) **do**
4: **if** $\exists \mathbf{S} \subseteq \mathbf{C} \setminus \{X\}$ with $T \perp\!\!\!\perp X \mid \mathbf{S}$ **then** remove X from $\mathbf{C}$
5: **return C**

neighbors of the target; the PC-of-PC search brings in co-parents; CI tests remove false positives.

Halfway: the CI column is filled (GS: a blanket per node; HITON: one target). The score-local corner remains: MMHC and BOSS.

3 Bringing in the score

Purpose. Combine local CI skeletons with scores, and search over orders.

3.1 MMHC: a hybrid bridge

Start at CI-local, then switch to score (Tsamardinos, Brown and Aliferis 2006). Run local parents-and-children discovery for every node with MMPC, the max-min sibling of HITON-PC (same grow/shrink, different admission rule); the union is a candidate skeleton. A score-based hill-climb then picks a DAG but may add edges only inside that skeleton:

$$\text{local PC per node (MMPC)} \;\rightarrow\; \text{union} = \text{skeleton} \;\rightarrow\; \text{score hill-climb} \;\rightarrow\; \text{one DAG}.$$

It combines a sound skeleton restriction with heuristic score-based orientation: the first time a score drives the graph.

3.2 BOSS: best order score search

Score-local (Andrews et al. 2023). The search state is a permutation π, not a graph. Each v's candidate parents are its *prefix*, the variables before it:

$$\text{pre}_\pi(v) = \{w : \pi.\text{index}(w) < \pi.\text{index}(v)\};$$

e.g. for $\pi = (A, C, B, D, E)$, $\text{pre}_\pi(B) = \{A, C\}$.

Scoring an order. The projection $\mathcal{G}_\pi$ picks each v's parents from its prefix by greedy grow/shrink, approximating

$$\text{pa}_\pi(v) \approx \underset{\mathbf{W} \subseteq \text{pre}_\pi(v)}{\arg\max}\ \text{score}_v(\mathbf{W}), \qquad \text{score}_v(\mathbf{W}) = \text{BIC}(X_v \mid X_{\mathbf{W}}),$$

a penalized log-likelihood (higher = better fit with fewer parameters). The order score sums the local terms, $\text{score}(\pi) = \sum_v \text{score}_v(\text{pa}_\pi(v))$. Grow–shrink trees (GSTs) cache repeated prefix scores; this is the key to BOSS's scaling. It relies on the local decomposition $\text{BIC}(\mathcal{G}, D) = \sum_{v \in \mathbf{V}} \text{BIC}(X_v \mid X_{\text{pa}_\mathcal{G}(v)})$: changing v's parents affects only one local term, which makes projection and best-move scoring efficient.

Algorithm 3 project($\mathbf{X}, \pi$) and best-move($\mathcal{T}, \pi, v$)

$E \leftarrow \emptyset$
for each v in π **do**
    $\mathbf{Z} \leftarrow \text{pre}_\pi(v)$; $\mathbf{W} \leftarrow \text{grow}(\mathbf{X}, v, \mathbf{Z})$; $\mathbf{W} \leftarrow \text{shrink}(\mathbf{X}, v, \mathbf{W})$
    add $w \to v$ to E for each $w \in \mathbf{W}$
return $\mathcal{G}_\pi = (\mathbf{V}, E)$

best $\leftarrow \mathcal{T}.\text{score}(\pi)$
for $i = 1, \ldots, |\pi|$ **do**
    $\pi' \leftarrow \pi.\text{move}(v, i)$
    if $\mathcal{T}.\text{score}(\pi') >$ *best* **then** *best* $\leftarrow \mathcal{T}.\text{score}(\pi')$; $\pi \leftarrow \pi'$
return π

Example 3.1 (Best-move)**.** Truth: the collider $A \to C \leftarrow B$. Move one variable to every position, re-project, and keep the move only if $\text{score}(\mathcal{G}_{\pi'}) > \text{score}(\mathcal{G}_\pi)$. With $\pi = (A, C, B)$, C precedes B, so C is a candidate parent of B; conditioning on the collider C couples A and B, so the projection includes an extra (spurious) A–B dependence: 3 edges. Moving C last, $\pi' = (A, B, C)$, gives the true collider with 2 edges. The independences represented are those of the true model with fewer parameters, so $\text{score}(\mathcal{G}_{\pi'}) > \text{score}(\mathcal{G}_\pi)$ and π' is kept.

Benchmarks. BOSS is designed for large, relatively dense models. In reported simulations it matches GRaSP (an earlier order-based search) in accuracy while scaling far better in runtime, compares favorably with fast GES, PC, DAGMA and LiNGAM, and has been demonstrated on resting-state fMRI with hundreds of cortical parcels (dense graphs, hundreds to 1000 variables).

4 Zoom out

Purpose. Choose among the methods and remember what none of them solve.

The filled map. Going local keeps the signal but shrinks the object; MMHC bridges CI and score; GSTs make the score-local corner scale.

What to use when.

- Sparse graph, reliable CI tests: GS (full graph) or HITON (one target).
- High-dimensional variable/feature selection: HITON's target blanket.
- Large, dense, score-friendly data: BOSS (or MMHC as a simpler hybrid).
- Hidden confounders: none of these; use FCI-type methods.
- Only one effect is needed, not the whole graph: effect-specific local discovery (Entner et al.; LDECC).

What none of these solve / limitations checklist.

- Causal sufficiency: GS, HITON, MMHC and BOSS all assume no hidden confounders; none replaces FCI.
- Faithfulness: separations must mirror independences (or score consistency must hold).
- Finite samples: weak effects and large conditioning sets hurt every method.
- Specification and tuning: the score family (BOSS needs an appropriate local likelihood) and the penalty (e.g. BIC) set the precision–recall balance between sparse and dense recovery.

Locality changes the cost of search, not the assumptions.

Comparison.

	GS	HITON	MMHC	GES	BOSS
Main signal	CI tests	CI tests	CI + score	BIC-like score	BIC-like score
Search object	blankets	target blanket	skeleton + DAG	CPDAGs	orders
Locality	Markov blankets	PC/co-parents	local skeleton	edge operators	prefixes
Output target	MEC graph	variables for T	one DAG	CPDAG	CPDAG
Main risk	CI error	target scope	heuristic orient.	many score calls	model spec.

Summary. One primitive, grow–shrink, on four local objects: GS (CI on Markov blankets), HITON (CI on one target's neighborhood), MMHC (a CI skeleton restricting a score hill-climb), BOSS (score on order prefixes, with GSTs caching repeated work). The same signals as before, made local: cheaper search, same assumptions.

5 Source coverage

The source deck has 46 physical pages (18 numbered slides plus appendix slides).

Topic	Physical pages
Map, axes, grow–shrink primitive	1–7
Markov blankets, GS, HITON	8–17
MMHC, BOSS, best-move, benchmarks	18–26
Zoom out: filled map, guidance, limitations, summary	27–35
Appendix: pseudocode, correctness, structure induction, comparison	36–46